

28	C	The range of wavelengths for visible light is 400 nm to 700 nm. Since $E = \frac{hc}{\lambda}$, the energies of these photons range from $\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(700 \times 10^{-9})(1.60 \times 10^{-19})} = 1.7759 \text{ eV to } \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(400 \times 10^{-9})(1.60 \times 10^{-19})} = 3.1078 \text{ eV} .$ Only 3 transitions will result in emissions of such photons: $6.12 - 4.28 = 1.84 \text{ eV}$ $6.81 - 4.28 = 2.53 \text{ eV}$ $7.02 - 4.28 = 2.74 \text{ eV}$
29	D	When each electron is accelerated through the electric field